

CSE5026: Introduction and Frontiers of Cognitive Science

Neural Basis of Cognition

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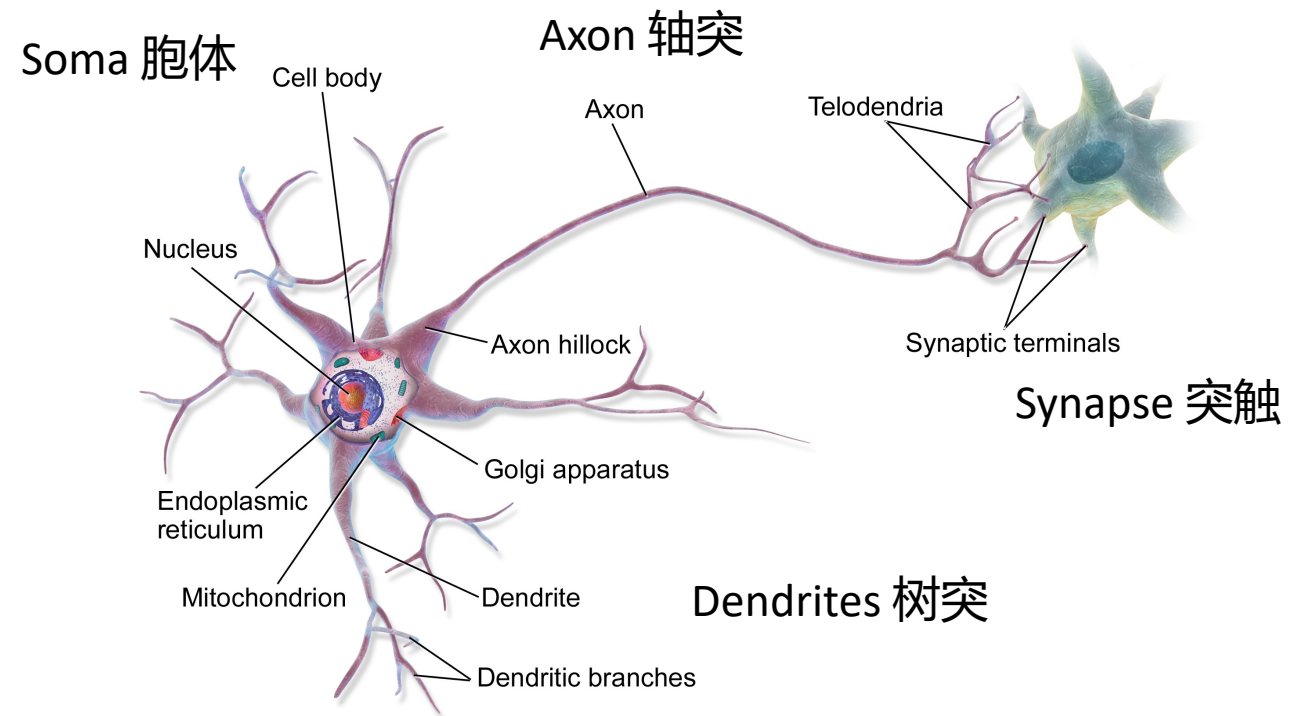
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Overview

- Neuron 神经元
- Resting Membrane Potential and Action Potential
静息膜电位与动作电位
- Leaky Integrate-and-Fire (LIF) Neurons 渗漏积分激发模型
- Brain

Neuron 神经元

- Neurons (神经元), or nerve cells (神经细胞), are the basic units of the nervous system (神经系统).
- A neuron typically has a soma / cell body (胞体), an axon (轴突), and dendrites (树突).

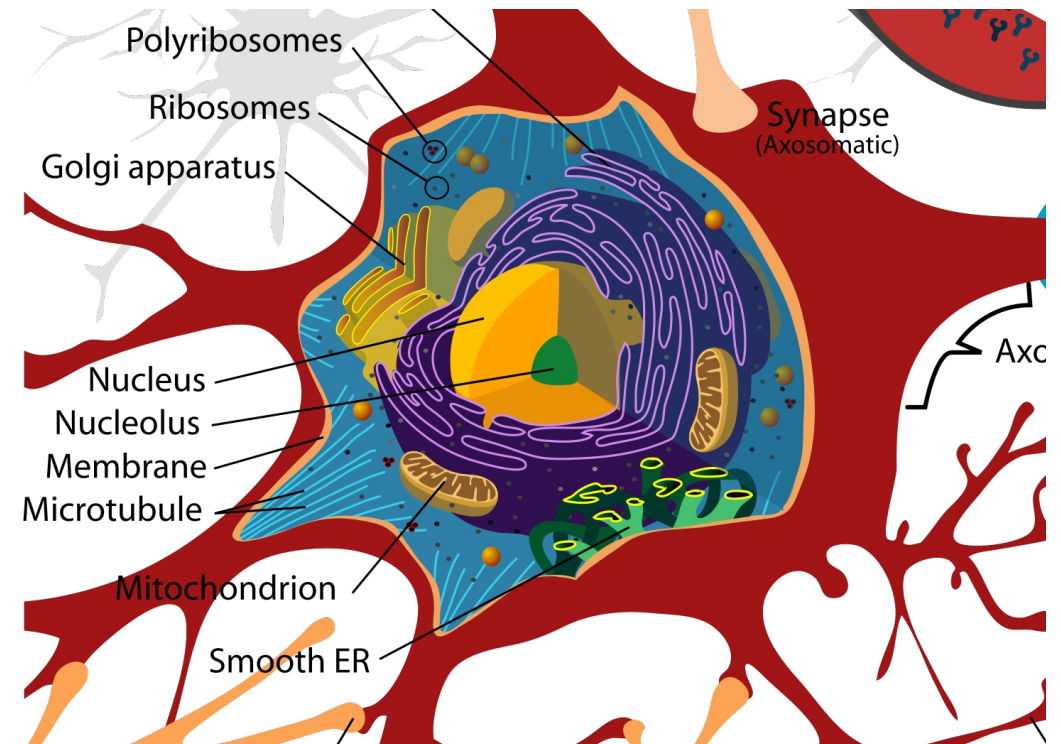


Source: <https://en.wikipedia.org/wiki/Neuron>

Soma / Cell Body 胞体

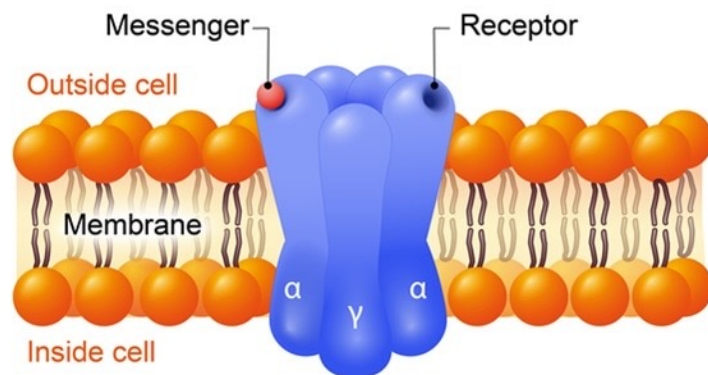
- Cell bodies occur in the brain (脑), spinal gray matter (脊髓灰质), and ganglia (神经节).
- Their diameters range from 5 to 150 μm .
- The soma is the neuron's metabolic and nutritional center.
- Its structure resembles that of other cells:
 - Nucleolus (核仁), membrane (细胞膜), cytoplasm (细胞质), and nucleus (细胞核).

Source: <https://en.wikipedia.org/wiki/Neuron>



Cell Membrane 细胞膜

- The neuronal membrane (神经细胞膜) is sensitive and excitable.
- Membrane proteins form receptors (受体) and ion channels (离子通道).
- Receptors bind specific neurotransmitters (神经递质).
- Acetylcholine (乙酰胆碱) or GABA (γ -氨基丁酸) binding changes membrane permeability and potential (膜电位), causing excitation (兴奋) or inhibition (抑制).



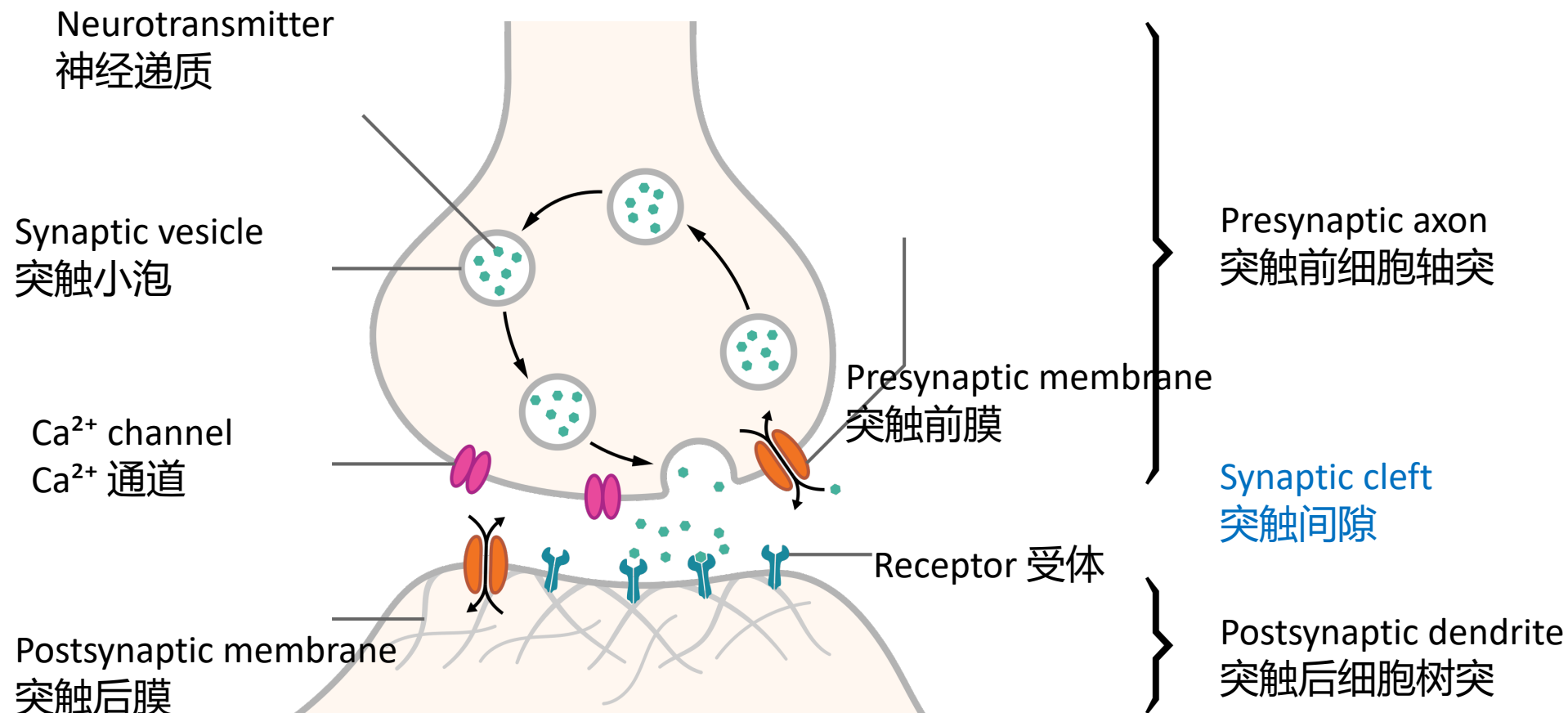
Synapse 突触

查尔斯·谢灵顿

- In 1897, Charles Sherrington (1857–1952) introduced the term synapse (突触) for the junction between two neurons.
- He proposed that neurons transmit information at these junctions.



Synapse Structure 突触结构

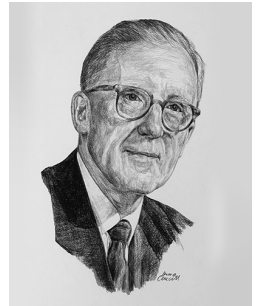


Source: <https://zh.wikipedia.org/zh-cn/突触>

Synaptic Transmission Process 突触传递 (brief)

- Electrical $\Rightarrow$ chemical $\Rightarrow$ electrical signaling
- Presynaptic excitation (突触前兴奋) **depolarizes** the presynaptic membrane (突触前膜去极化), allowing Ca^{2+} to enter.
- Neurotransmitters (神经递质) enter the synaptic cleft (突触间隙), diffuse across it, and bind postsynaptic receptors (突触后受体).
- Ion channels (离子通道) open, causing postsynaptic **depolarization** (去极化) or **hyperpolarization** (超极化).

Hebbian Learning at Synapses



- In *The Organization of Behavior* (1949), Canadian psychologist Donald O. Hebb proposed that synaptic connections (突触连接) can **change**.
- This hypothesis became known as Hebbian learning (赫布学习).
- Hebb's proposal (paraphrased): repeated coactivation links neurons so that one helps activate the other. If neuron A repeatedly activates B, their synaptic connection grows stronger.
- Synaptic plasticity (突触可塑性): “Cells that fire together, wire together.”

Hebbian Learning and Neural Networks

- In artificial neural networks (人工神经网络), changes in connection weights represent changes in synaptic transmission (突触传递).
- Hebbian learning rule:

$$w_{ij} = x_i x_j$$

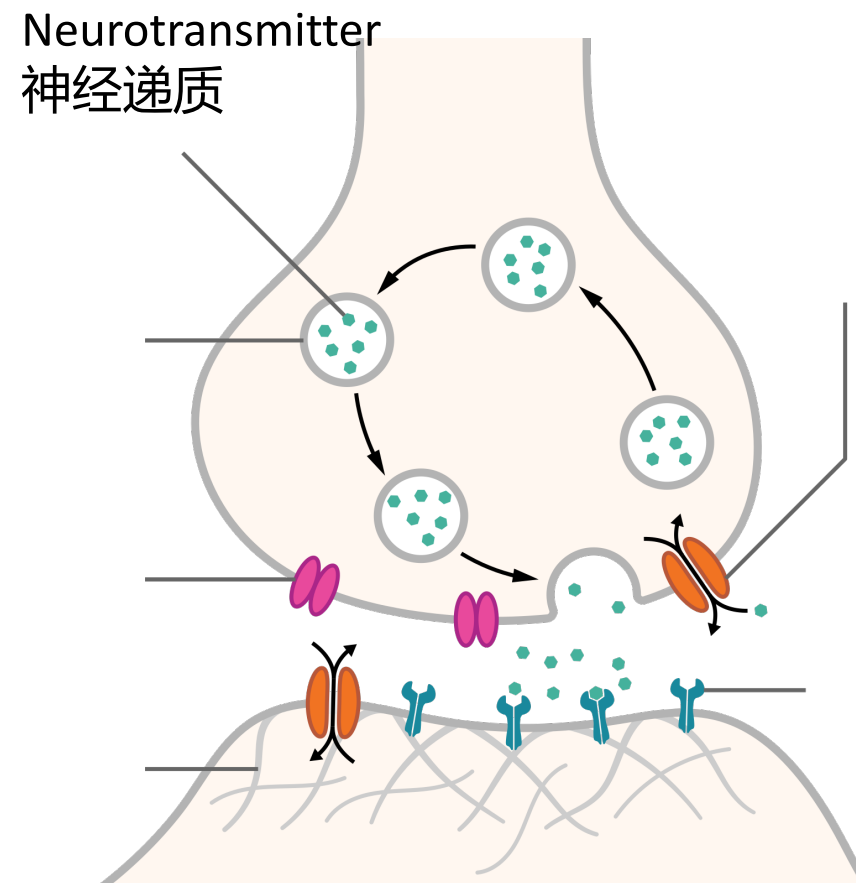
- w_{ij} is the weight from neuron j to neuron i . x_i is the input to neuron i , with $x_i \in \{0, 1\}$. Each training sample changes the weights.
- An expression over multiple samples:

$$w_{ij} = \frac{1}{p} \sum_{k=1}^p x_i^{(k)} x_j^{(k)}$$

p is the number of training samples.
 $x_i^{(k)}$ is the k th input to neuron i .

Neurotransmitters 神经递质

- Neurotransmitters (神经递质) are chemicals released by presynaptic terminals (突触前末梢). They bind receptors on postsynaptic neurons (突触后神经元) or effector cells (效应器细胞) to transmit information.
- Examples include:
 - Acetylcholine (乙酰胆碱)
 - Catecholamines (儿茶酚胺): norepinephrine (去甲肾上腺素) and dopamine (多巴胺)
 - Serotonin / 5-HT (5-羟色胺 / 血清素)

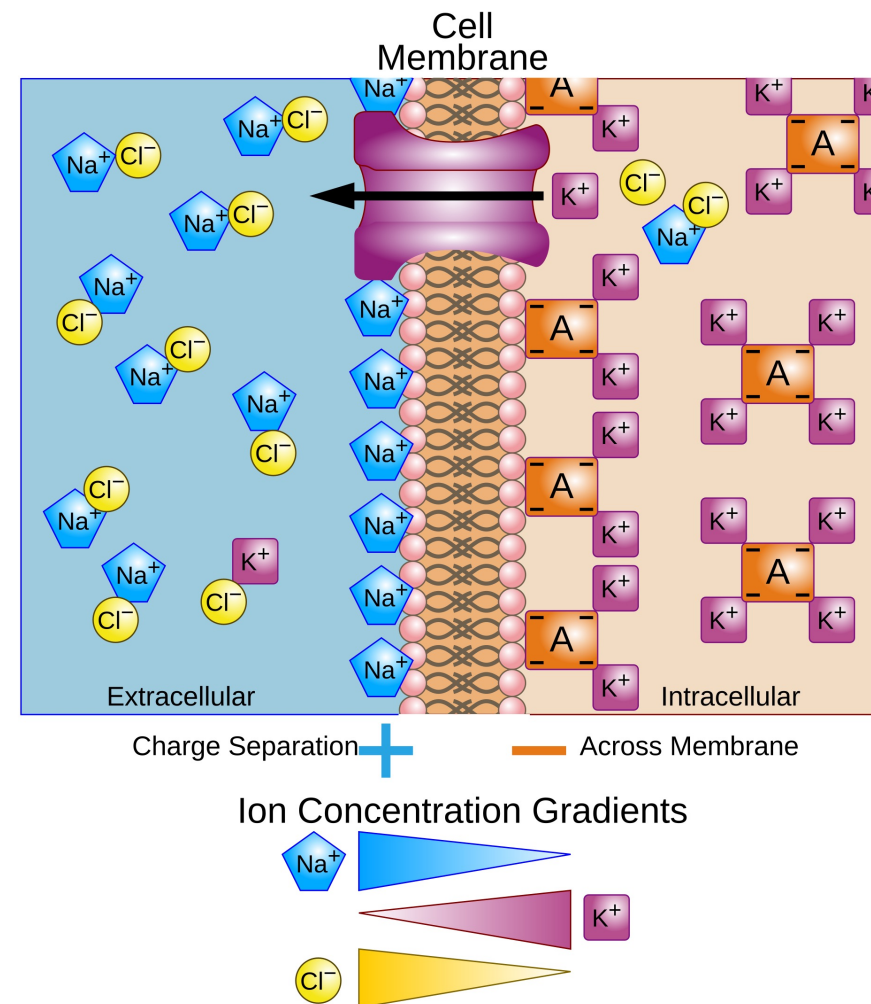


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- **Resting Membrane Potential Action Potential**
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- Brain 大脑

Resting Membrane Potential 静息膜电位

- At rest, neurons (神经元) have a membrane potential of about **-70 mV**, with the **inside negative relative to the outside**.
- Ion concentrations differ across the membrane:
 钠离子 氯离子 outside, and more 钾离子 inside.
 - The sodium–potassium pump (钠钾泵) uses one ATP to move 3 Na^+ out and 2 K^+ in.



Source: https://en.wikipedia.org/wiki/Membrane_potential

Hodgkin–Huxley Experiments

- British physiologists A. L. Hodgkin and A. F. Huxley inserted glass microelectrodes (玻璃微电极) into squid giant axons (乌贼巨轴突) to record membrane potentials inside cells.
- Giant axons are about 1 mm in diameter, roughly 100 times larger than typical axons, making experiments easier.
- Voltage clamp (电压钳) technique



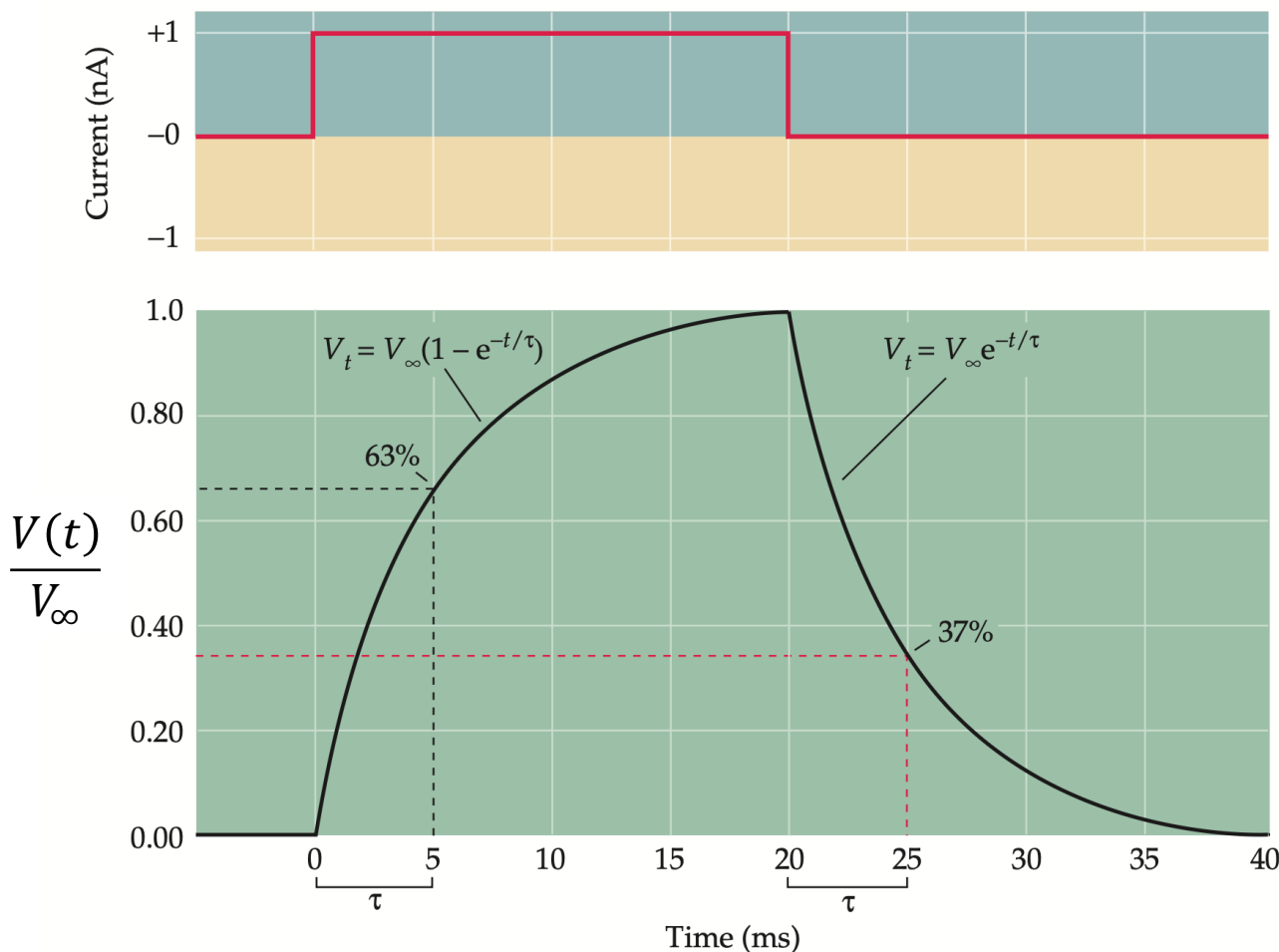
Source: https://en.wikipedia.org/wiki/Squid_giant_axon



HH's experiment:
Insert a current (电流) to the inside of neuron (axon), and watch how the membrane potential changes in relative to resting potential.

Hodgkin–Huxley Experiments

- Let $V(t)$ be membrane potential in relative to resting potential $V_{\text{rest}} \approx -70\text{mV}$



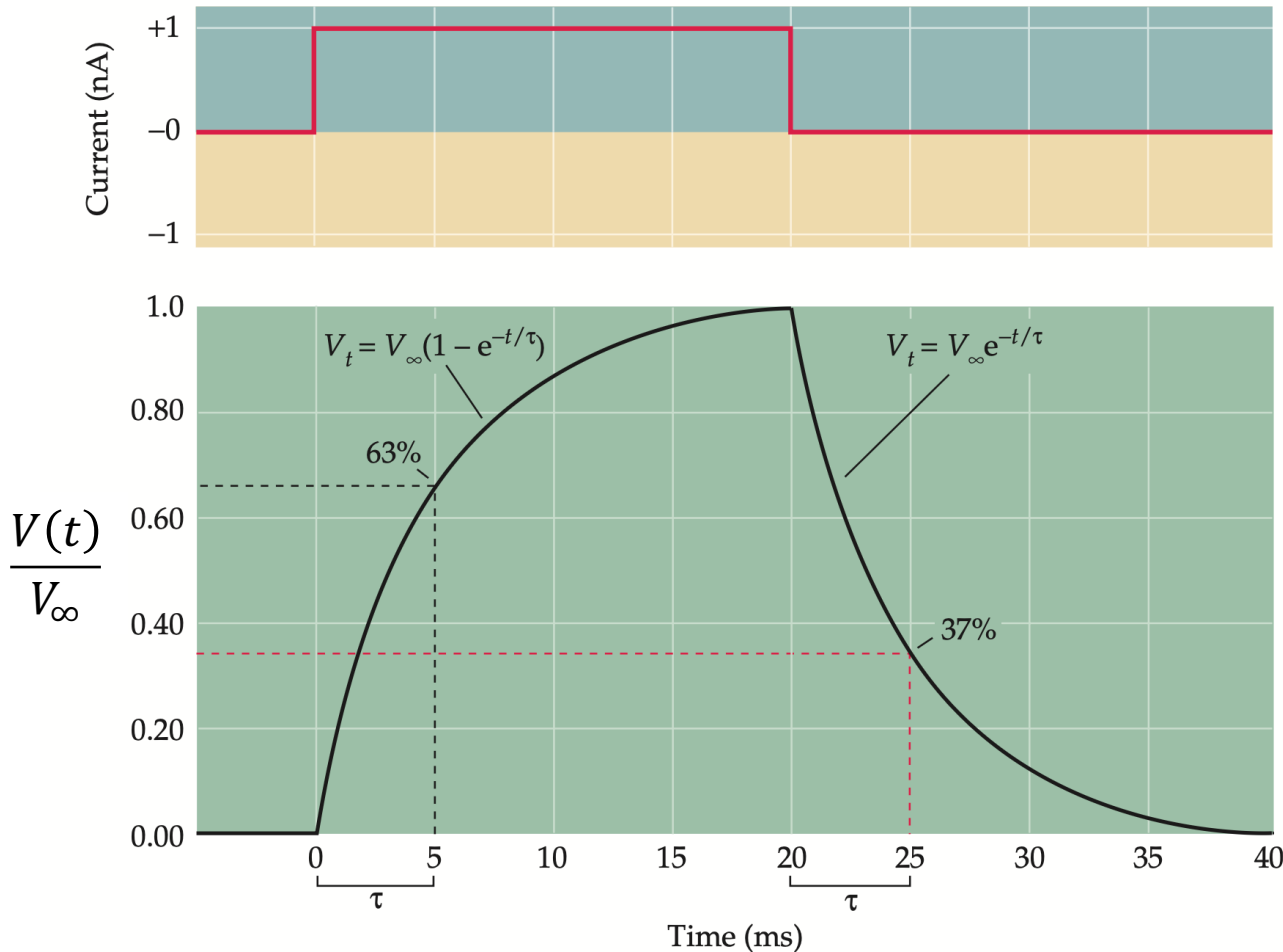
Membrane acts as both a resistor (电阻) and a capacitor (电容).

The current I has two parts:

$$I = \underbrace{C \frac{dV(t)}{dt}}_{\text{charges the membrane}} + \underbrace{\frac{V}{R}}_{\text{go through the membrane (Ohm's law)}}$$

Hodgkin–Huxley Experiments

- $V(t)$ expressed as an ODE



Moving terms in previous equation:

$$I = RC \frac{dV(t)}{dt} = IR - V(t)$$

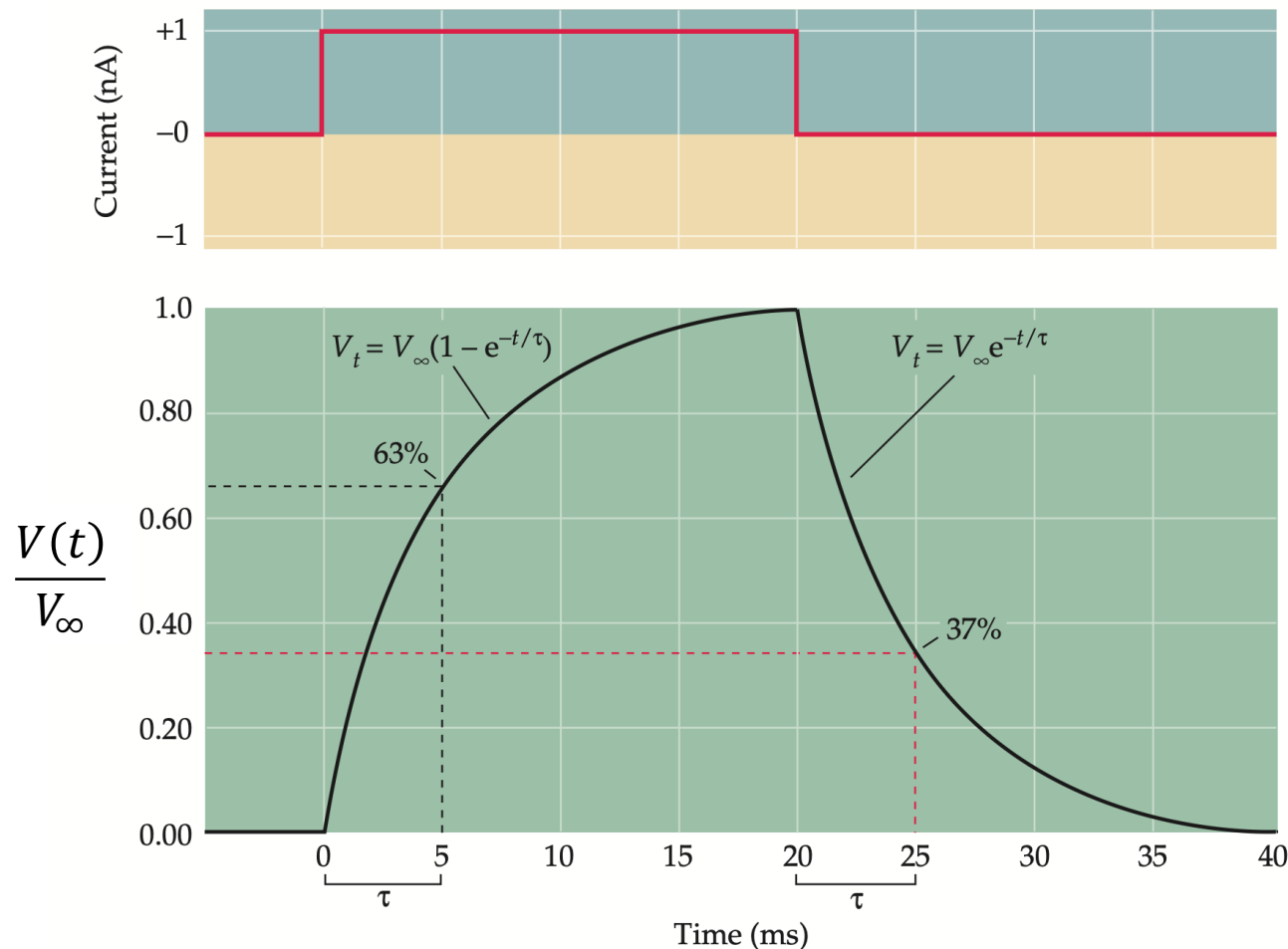
When $I = I_0$, i.e., there is a constant current injection, then $I_0 R = V_\infty$ is the final stabilized voltage

Let $\tau = RC$ be the *membrane time constant*, then the ODE is:

$$\frac{dV(t)}{dt} = -\frac{V(t)}{\tau} + \frac{V_\infty}{\tau}$$

Hodgkin–Huxley Experiments

- Solving the ODE



It can be solved analytically:

$$V(t) = (1 - e^{-t/\tau})V_\infty$$

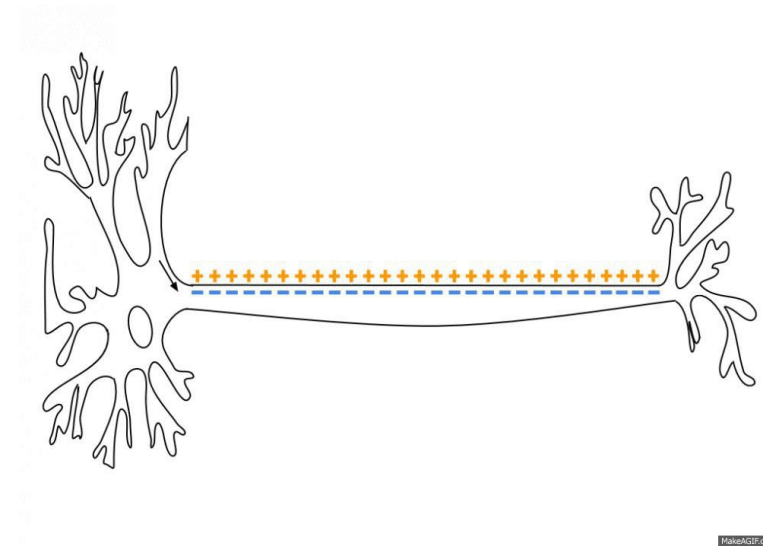
Meaning of τ :

when $t = \tau$, $V(t) = V_\infty \left(1 - \frac{1}{e}\right) \approx 0.63V_\infty$, the time for membrane potential to reach $0.63V_\infty$.

When the current injection is gone, $I = 0$, and $V(t) = e^{-t/\tau} \cdot V_\infty$

Action Potential 动作电位

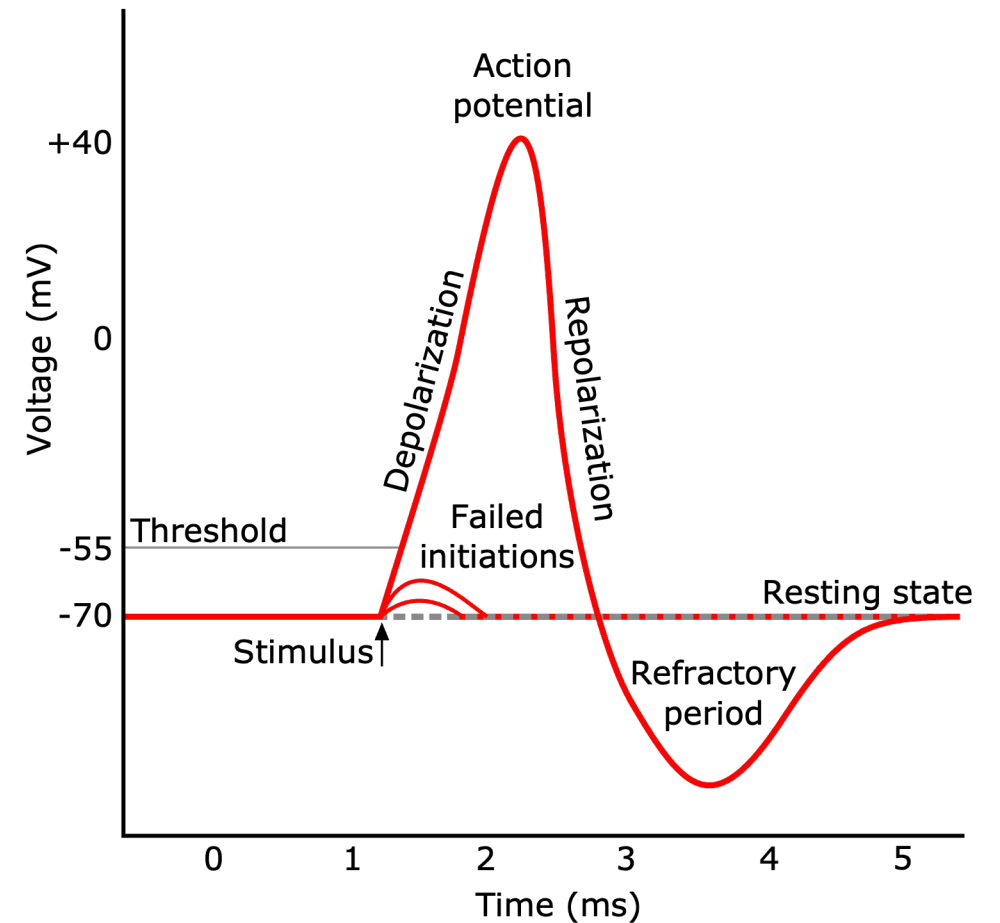
- Action potential (动作电位)
- A stimulus triggers a rapid reversal and restoration of the voltage across a cell membrane at rest.
- All-or-none principle (全有全无律)
- Propagation without decrement (不衰减传导)



Source: <https://zh.wikipedia.org/zh-cn/动作电位>

Action Potential 动作电位

- An action potential (动作电位) rapidly makes the inside of a local membrane region **positive** relative to the outside.
- A stimulus below the threshold potential (阈值电位) does not trigger an action potential.
- Once the stimulus reaches threshold, an action potential rapidly rises to its peak, then returns to the resting membrane potential (静息膜电位).



Source: <https://zh.wikipedia.org/zh-cn/动作电位>

Action Potential 动作电位

- All-or-none principle (全有或全无)
- This region acts like a battery and drives neighboring membrane above threshold (阈值), triggering another action potential.
- The signal propagates over long distances without decrement (不衰减传导). At axon terminals (轴突末梢), a smaller diameter may reduce its amplitude.

Overview

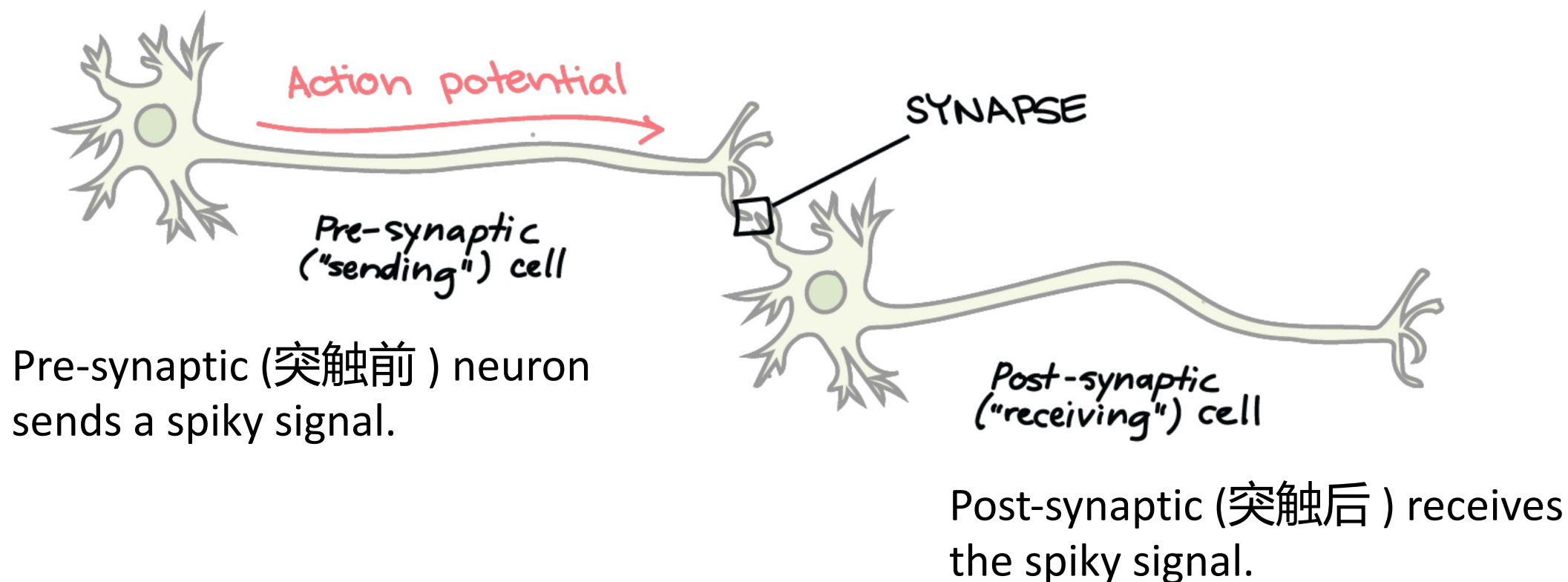
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Neuromorphic Computing

- Neuromorphic computing designs computing systems inspired by how biological nervous systems process information.
- A popular paradigm: neurons maintain internal states, communicate through **spikes**, and interact through weighted synapses
- Spiking neural networks (SNN) run in software or on specialized hardware

Spiking Neural Networks

- Spiking neural networks (脉冲神经网络)



When and how do neurons spike?

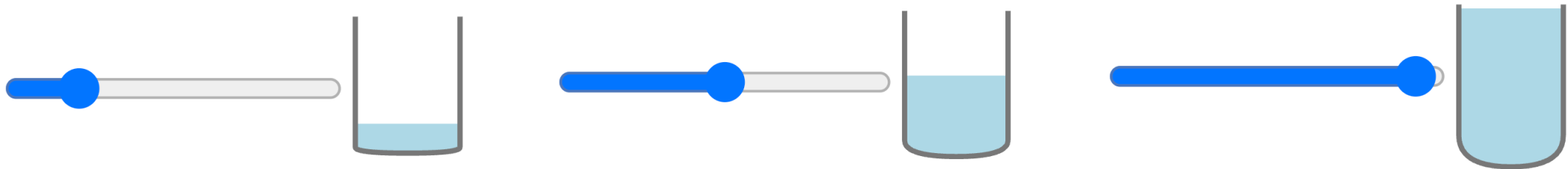
- A simple set of rules determine when neurons spike.
- A neuron *builds up* potential energy (势能) when it receives pre-synaptic neuron spikes.
- When the potential energy exceeds a threshold, the neuron spikes.
- **Leaky Integrate-and-Fire (LIF)** neurons 渗漏积分激发 模型。

Leaky Integrate-and-Fire (LIF) Neurons

- The post-synaptic neuron adds up (**integrates**) the inputs from all the pre-synaptic neurons but its value decreases (**leaks**) over time.
- If the sum of those inputs reaches a minimum threshold, it **fires**.
- Not a precise simulation but captures the important functional aspects of biological neurons.
- (*“All models are wrong but some are useful”*)

LIF: A Leaky Bucket Analogy

- Imagine the neuron as a bucket with a bottom made of rubber (橡胶).
- The more water we add, the higher “potential” the neuron has.



- Full bucket => the neuron is close to firing.

Source: <https://soney.github.io/snn-from-scratch/chapters/03 - Basic Neuron Models.html>

LIF: A Leaky Bucket Analogy

- Imagine we poke a small hole in the rubber bottom, then the water will slowly leak out of the role.
- If we add more water, the hole will expand and let out water faster.
- An analogy to biological neurons, whose the membrane potential decreases over time without additional inputs.



Source: <https://soney.github.io/snn-from-scratch/chapters/03 - Basic Neuron Models.html>

LIF: Mathematical Notations

- Refer to the potential as v (for voltage), which changes over time as a function of time t : $v(t)$
- Derivative $v'(t)$: the rate of change of $v(t)$ at time t . (the speed at which water leaks out of bucket)
 - $v'(t) < 0$: we are losing water; $v'(t) > 0$: adding water.
- $v'(t)$ is determined by a few factors:
 - $I(t)$: the amount of incoming water at time t
 - $v(t)$: current amount of water (water leaks out faster if the bucket is more full)
 - τ : how the hole grows in relation to the amount of water (how “stretchy” the water is)

LIF: Mathematical Notations

- Use a formula:

$$v'(t) = \frac{1}{\tau} I(t) - \frac{1}{\tau} v(t)$$

- If $I(t) = 0$: no incoming water, then

$$v'(t) = -\frac{1}{\tau} v(t)$$

- The water leaks out
slower for larger values of τ ;
faster for smaller values.

$\tau = 500$:



$\tau = 1000$:



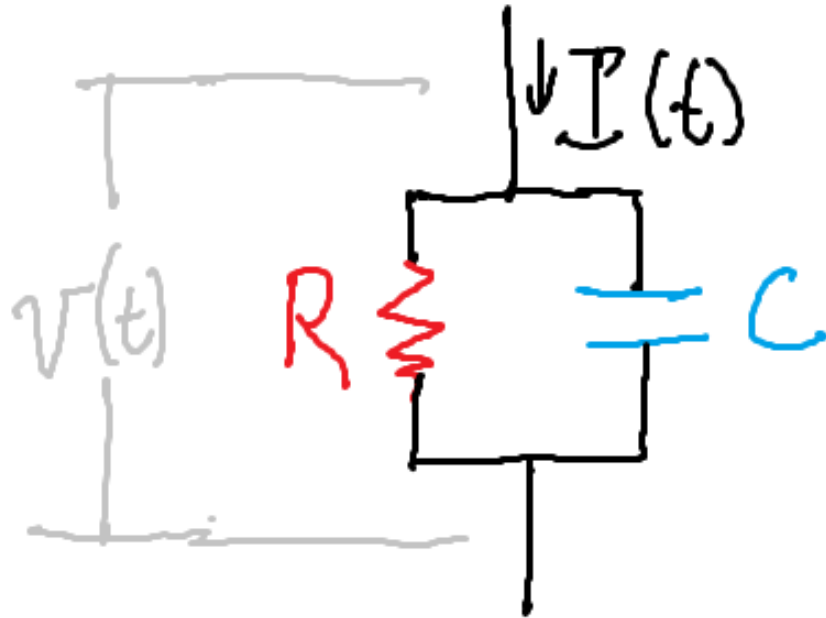
$\tau = 3000$:



Source: <https://soney.github.io/snn-from-scratch/chapters/03 - Basic Neuron Models.html>

Resistor-Capacitor (RC) circuit model

- A more “standard” way of introducing LIF model



R : resistance of a resistor;

C : capacitance of a capacitor

$I(t) = I_R(t) + I_C(t)$: the amount of current that flows through the resistor and capacitor

Ohm's law: $I_R(t) = \frac{v(t)}{R}$

For the capacitor: $I_C(t) = C v'(t)$

Source: <https://soney.github.io/snn-from-scratch/chapters/03 - Basic Neuron Models.html>

Resistor-Capacitor (RC) circuit model (cont.)

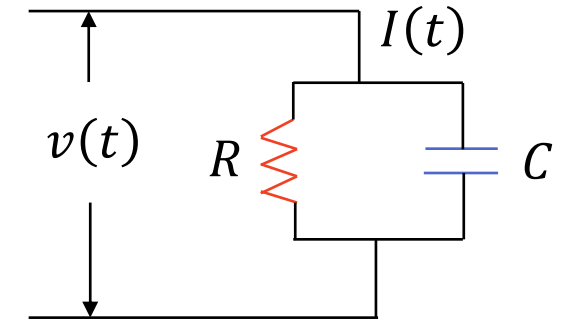
- Thereby, $I(t) = I_R(t) + I_C(t) = \frac{v(t)}{R} + C v'(t)$

- $\Rightarrow v'(t) = \frac{1}{C} I(t) - \frac{1}{RC} v(t)$

- Let $I_2(t) = RI(t)$, and $\tau_{RC} = RC$, then: $v'(t) = \frac{1}{\tau_{RC}} I_2(t) - \frac{1}{\tau_{RC}} v(t)$

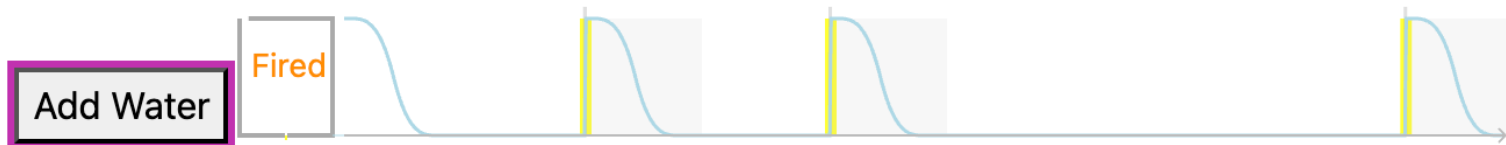
- Recall the previous generic formula:

- $v'(t) = \frac{1}{\tau} I(t) - \frac{1}{\tau} v(t)$. Now τ is replaced with τ_{RC} with a physical meaning.



LIF: Firing

- Once the potential reaches a firing threshold v_{th} , the neuron **fires** (triggers an action potential 动作电位).
- Threshold: the neuron will only fire when $v(t) \geq v_{th}$.
- After firing, then neuron enters a “refractory period” (不应期) where its potential resets and it cannot fire.



Source: <https://soney.github.io/snn-from-scratch/chapters/03 - Basic Neuron Models.html>

First-Order Approximation of LIF

- In continuous time, $v'(t) = \frac{1}{\tau_{RC}}I(t) - \frac{1}{\tau_{RC}}v(t)$ * we remove the subscript in the current term $I_2(t)$
- Using Euler's approximation method, given $v(t)$ and its rate of change $v'(t)$, we can estimate $v(t + \Delta t)$: $v(t + \Delta t) \approx v(t) + v'(t)\Delta t$
- Substitute $v'(t)$:

$$v(t + \Delta t) \approx v(t) + \left(\frac{1}{\tau_{RC}}I(t) - \frac{1}{\tau_{RC}}v(t) \right) \Delta t$$

$$v(t + \Delta t) \approx v(t) \left(1 - \frac{\Delta t}{\tau_{RC}} \right) + \frac{\Delta t}{\tau_{RC}}I(t)$$

Discrete Time Simulation

- Let $v[t]$ be an approximation of $v(t)$: $v[t] \approx v\left(\frac{1}{T_{step}}t\right)$,
where T_{step} is the minimal time step (small enough for getting an accurate Euler's approximation)
- Then we have the discrete form: $v[t + 1] \approx v[t] \left(1 - \frac{T_{step}}{\tau_{RC}}\right) + \frac{T_{step}}{\tau_{RC}} I(t)$
- which can be implemented iteratively:
- $v[0] = v_{init}$
- $v[t] = v[t - 1] \left(1 - \frac{T_{step}}{\tau_{RC}}\right) + \frac{T_{step}}{\tau_{RC}} I(t)$
* easier to implement by replacing $I(t - 1)$ with $I(t)$

Python Implementation of LIF

Source: <https://soney.github.io/snn-from-scratch/chapters/04 - First Order LI.html>

```
class FirstOrderLI: # First Order Leaky Integrate
    def __init__(self, tau_rc=0.2, v_init=0): # Default values for tau_rc and v_init
        self.tau_rc = tau_rc # Set instance variables
        self.v = v_init

    def step(self, I, t_step): # Advance one time step (input I and time step size t_step)
        self.v =
```

$$v[t] = v[t - 1] \left(1 - \frac{T_{step}}{\tau_{RC}}\right) + \frac{T_{step}}{\tau_{RC}} I(t)$$

- The potential $v[t]$ is stored in `self.v`
- Every time `step()` is called, the LIF model moves forward by one time step and updates `self.v`

Implementing Firing

- To implement the firing mechanism, we need to set the threshold potential and the refractory period.
- v_{th} : if the neuron's potential reaches this threshold, it will fire.
- τ_{ref} : after the neuron fires, it needs to wait this long until it “accepts” input again.
- Represent firing as the neuron's “output”
 - By default the output will be 0
 - When firing, the output will be $1/T_{step}$ (so that the total energy within one time step will be the same regardless of step length)

Python Implementation of LIF firing

Source: <https://soney.github.io/snn-from-scratch/chapters/04 - First Order LI.html>

```
class FirstOrderLIF: # First Order Leaky Integrate and Fire
    def __init__(self, tau_rc=0.2, tau_ref=0.002, v_init=0, v_th=1):
        self.tau_rc = tau_rc # Potential decay time constant
        self.v = v_init # Potential value
        self.v_th = v_th # Firing threshold
        self.tau_ref = tau_ref # Refractory period time constant

        self.output = 0 # Current output value
        self.refractory_time = 0 # Current refractory period time (how long until the neuron can fire again)

    def step(self, I, t_step): # Advance one time step (input I and time step size t_step)
        self.refractory_time -= t_step # Subtract the amount of time that passed from our refractory time

        if self.refractory_time < 0: # If the neuron is not in its refractory period
            self.v = self.v - (self.v - 0) * self.tau_rc * t_step # Integrate input and leak
            if self.v > self.v_th: # If the potential is above the threshold
                self.refractory_time = self.tau_ref # Enter the refractory period
                self.output = 1 # Emit a spike
                self.v = 0 # Reset the potential
            else: # If the potential is below the threshold
                self.output = 0 # Do not fire

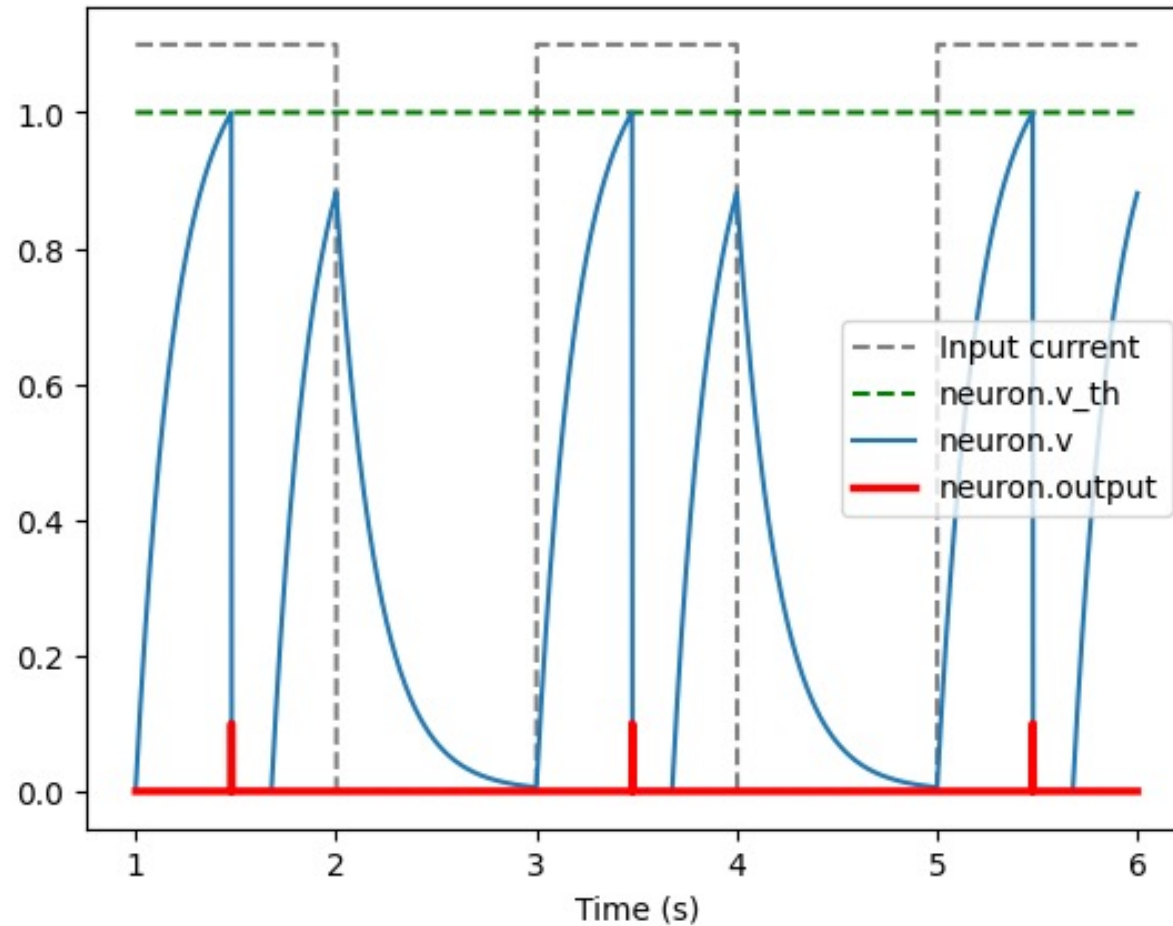
        return self.output
```

→ outside refractory time

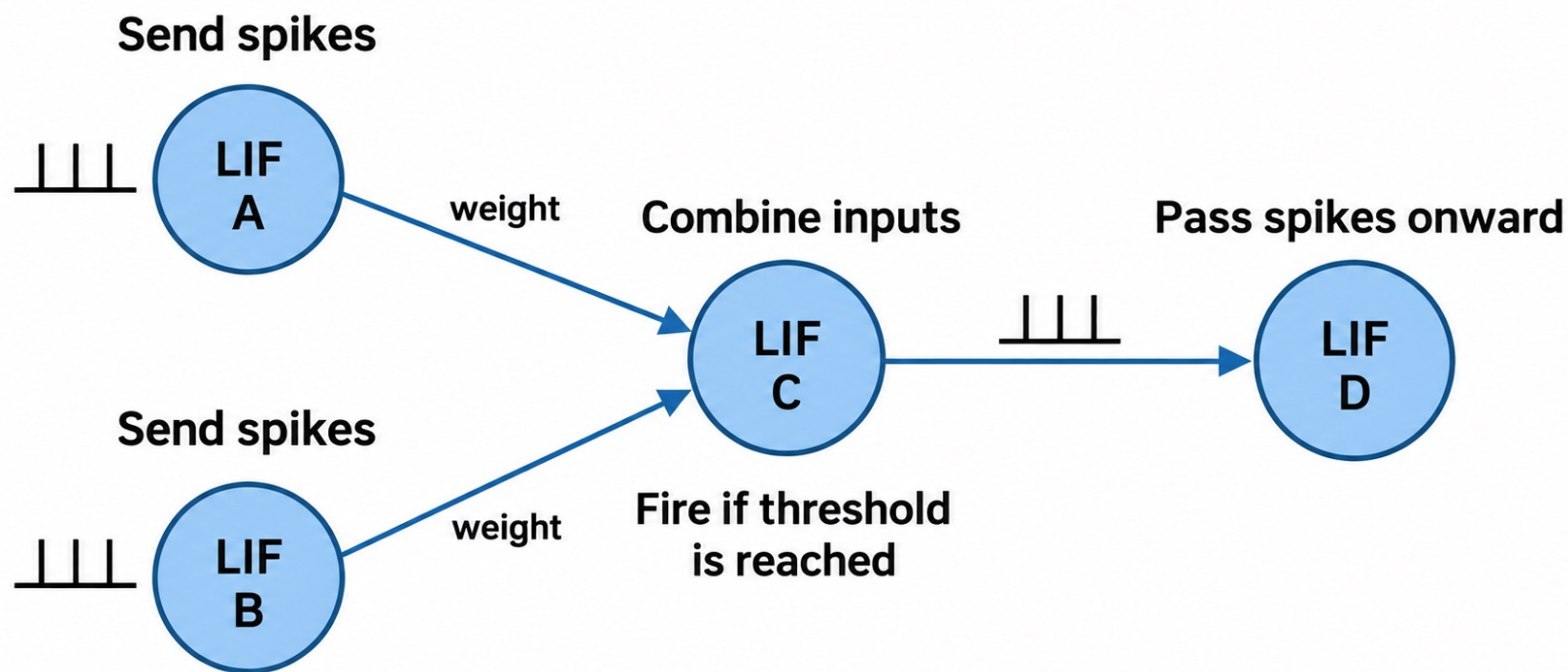
} fire and enter refractory time

Firing output

Source: <https://soney.github.io/snn-from-scratch/chapters/04 - First Order LI.html>



Connect LIF neurons to form SNNs



Synapses

A spike creates an input that fades over time.

Weights

Weights scale each connection's input.

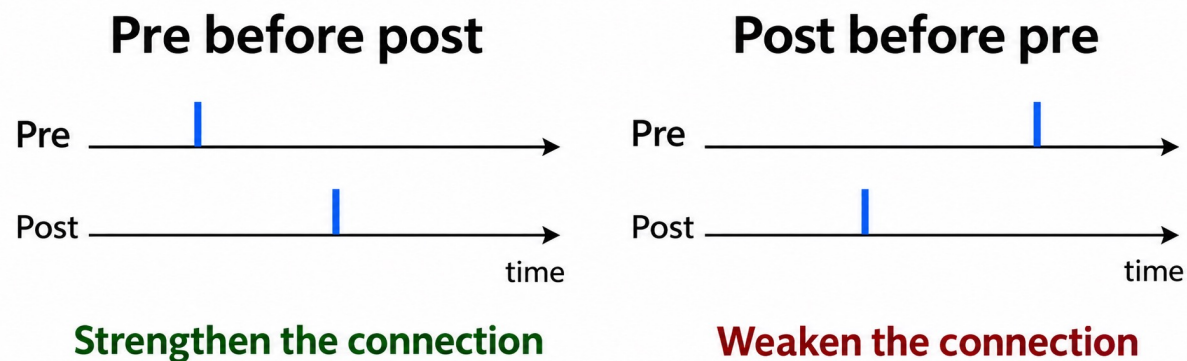
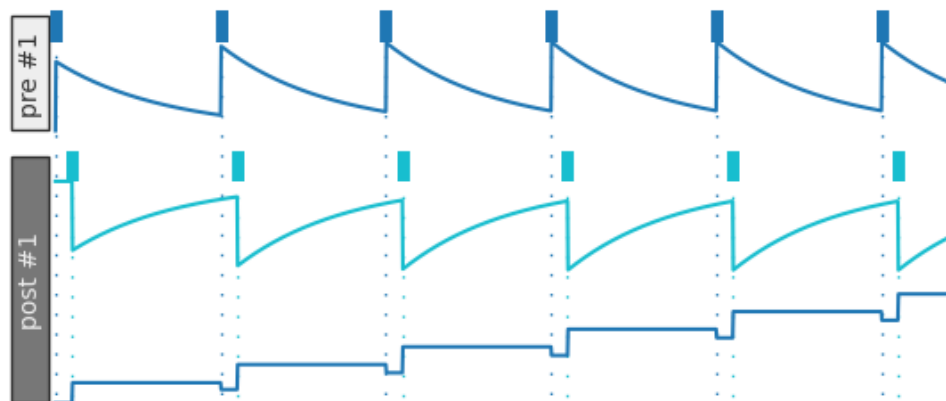
Network activity

Each neuron combines inputs, leaks, and fires at threshold.

STDP: Learning from spike timing

Spike-Timing-Dependent Plasticity (STDP): A learning rule based on spike order and time difference,

implementing **Hebbian learning**: “Cells that fire together, wire together.”

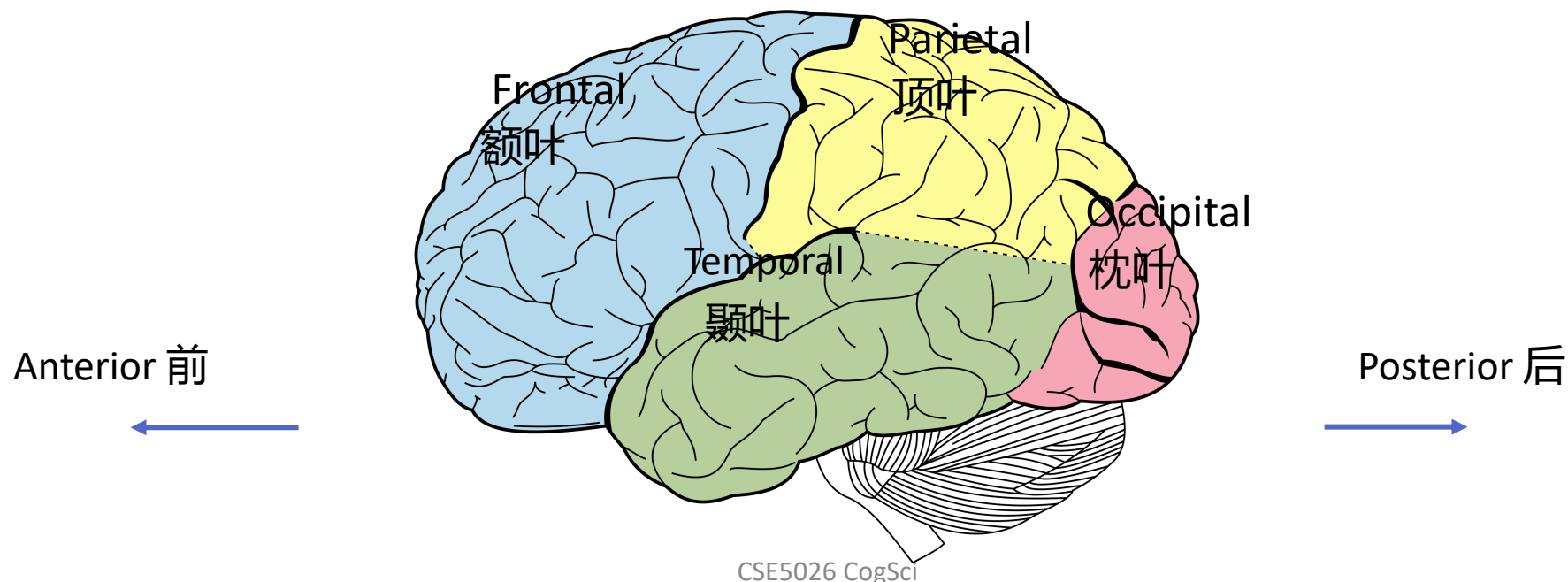


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- **Brain 大脑**

Structure of the Human Brain 人脑结构

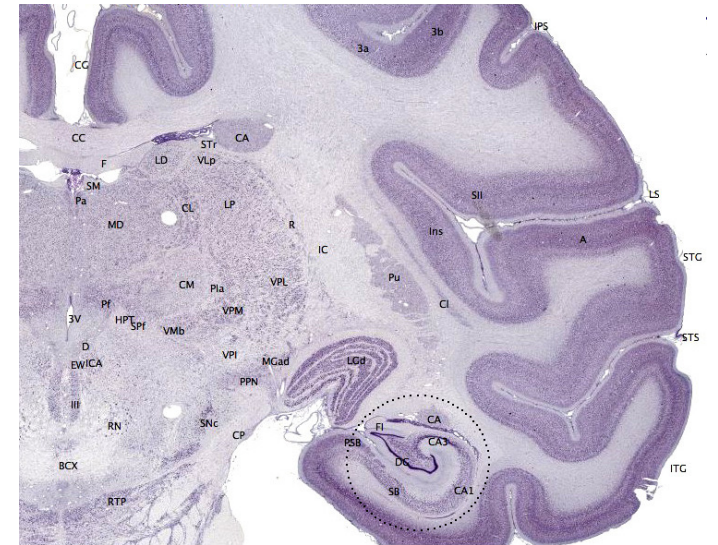
- The brain includes the cerebrum (大脑), cerebellum (小脑), brainstem (脑干), and diencephalon (间脑).
- The cerebrum has two hemispheres (大脑半球). Each has four lobes (脑叶): frontal (额叶), parietal (顶叶), temporal (颞叶), and occipital (枕叶).



Cerebral Cortex 大脑皮层 / 皮质

- The cerebral cortex (大脑皮层) consists of gray matter (灰质). It is a relatively recent structure in evolution and supports higher brain functions.
- It has inward folds, sulci (沟), and outward ridges, gyri (回). Its thickness is about 2–4 mm.
- It contains neurons (神经元) on the order of 10^{10} , or tens of billions.
- The neocortex (新皮层), unique to mammals, has six layers numbered I–VI from the surface inward.

Cortex (皮层) in dark purple



Source: <https://zh.wikipedia.org/zh-cn/大脑皮质>

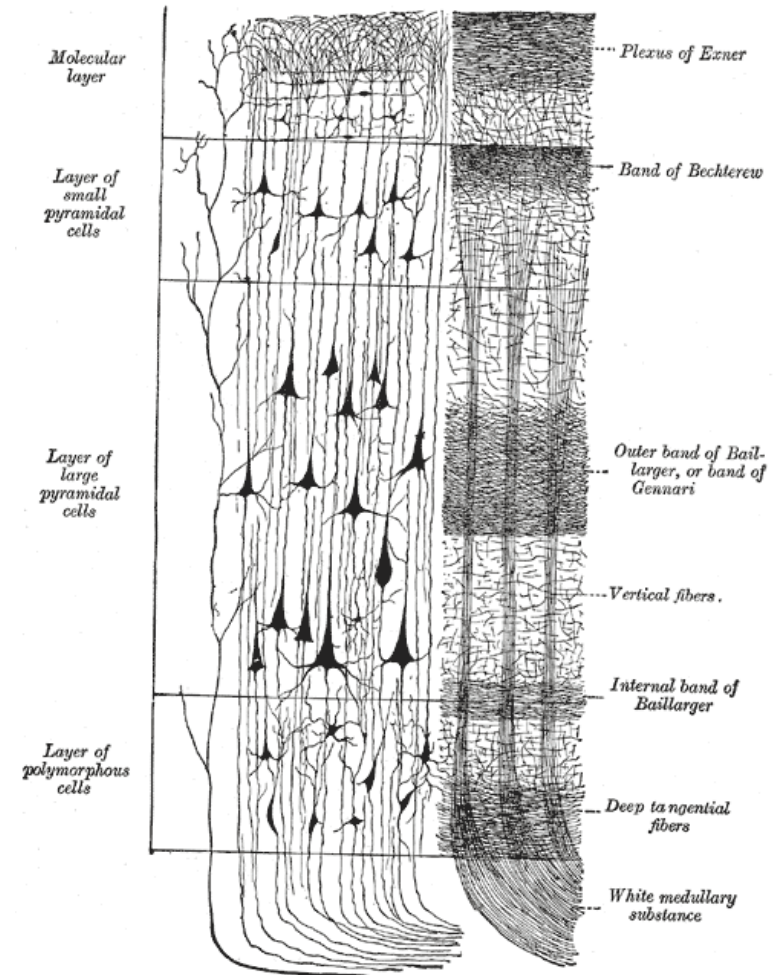
Brain Energy Use 大脑能耗

- The brain (大脑) is the body's most energy-demanding organ.
 - It accounts for about 2% of body mass, 20% of energy use, and 25% of oxygen demand.
- Cortical gray matter (皮层灰质) uses about 3 W. Glucose uptake suggests about 17–20 W for the whole brain, with roughly 9 W dissipated as heat.
- This power demand is small compared with that of computers.

Source: <https://swarma.org/?p=28187>

Layers of the Cerebral Cortex 皮质分层

- Cortical layers, from the surface inward:
- I. Molecular layer (分子层): few neurons, with small cell bodies.
- II. External granular layer (外颗粒层)
- III. External pyramidal layer (外锥体细胞层): relatively thick. Apical dendrites (顶树突) reach layer I, and axons extend into deeper layers.
- IV. Internal granular layer (内颗粒层)
- V. Internal pyramidal / ganglionic layer (内锥体细胞层 / 节细胞层)
- VI. Multiform / polymorphous layer (多形细胞层)



Functions of the Cortical Lobes 脑叶功能

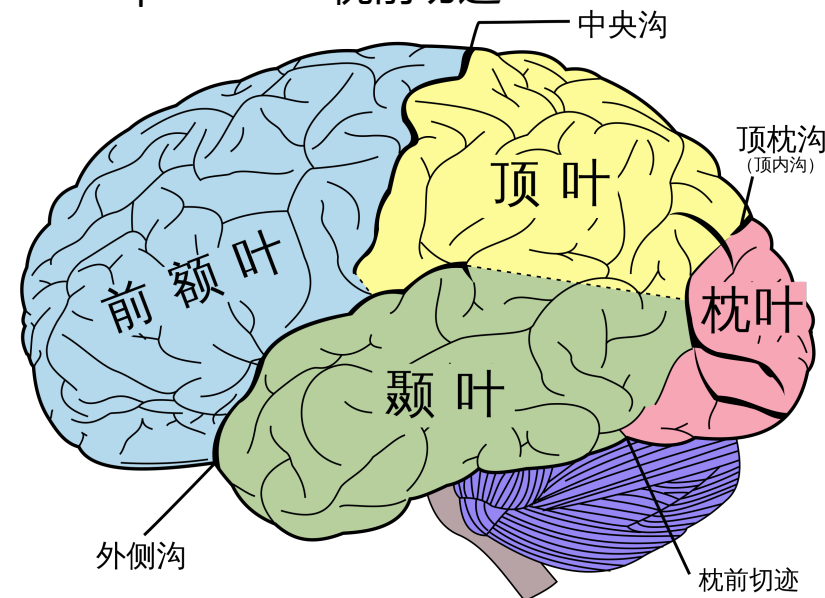
- Different lobes contribute to different functions.
- Frontal lobe (额叶): higher cognition, including learning, language, decision-making, abstract thought, and emotion.
- Parietal lobe (顶叶): somatosensation (躯体感觉) and integration of spatial, visual, and bodily information.
- Temporal lobe (颞叶): hearing, smell, higher visual processing, left-right discrimination, and long-term memory (长期记忆).
- Occipital lobe (枕叶): visual processing (视觉处理).

Central sulcus 中央沟

Lateral sulcus 外侧沟

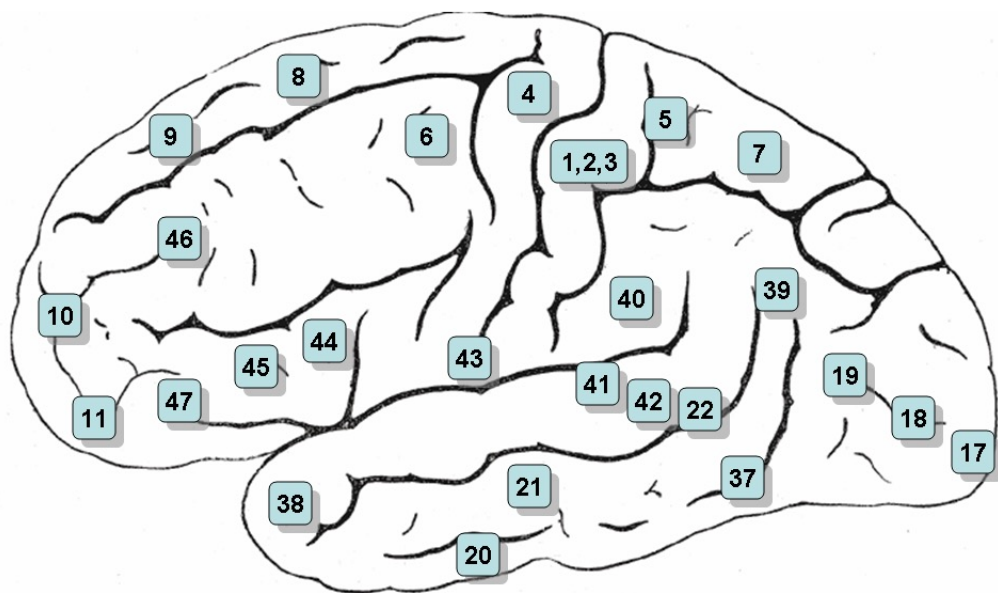
Parieto-occipital sulcus 顶枕沟

Preoccipital notch 枕前切迹

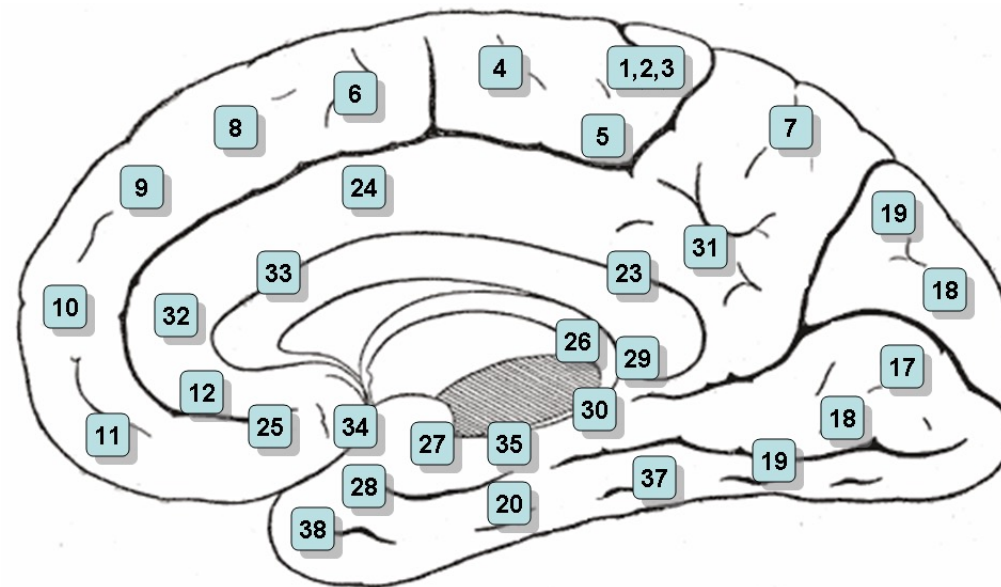


Brodmann Areas 布罗德曼分区

- German neurologist K. Brodmann (1868–1918) mapped 52 areas of the cerebral cortex (大脑皮层) by neuronal organization in stained brain tissue.



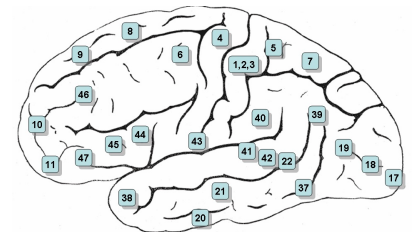
Lateral surface 外侧面



Medial surface 内侧面

Brodmann Areas 布罗德曼分区

- The cortex has three broad functional divisions: sensory cortex (感觉皮层), motor cortex (运动皮层), and association cortex (联合皮层).
- Sensory areas: visual (视觉, 17), auditory (听觉, 41–42), somatosensory (躯体感觉, 1–3), gustatory (味觉, 43), and olfactory (嗅觉, 28).
- Motor areas: primary motor cortex (初级运动皮层, 4), premotor cortex (运动前区), and supplementary motor area (辅助运动区, 6).
- Association cortex integrates sensory information and passes it to motor areas to guide behavior. Its mapping onto Brodmann areas is more complex.



Electrical Activity of the Cortex 皮层电活动

- Cortical neurons (皮层神经元) produce ongoing rhythmic voltage changes known as spontaneous brain electrical activity (自发脑电活动).
- Brain waves (脑电波) can be classified by frequency:
- Delta (δ 波): 0.5–3 Hz. Theta (θ 波): 4–7 Hz. Alpha (α 波): 8–13 Hz. Beta (β 波): above 14 Hz.
- Slow waves (慢波) tend to have larger amplitudes, and fast waves (快波) smaller amplitudes.